

Multivariate Probability Distributions Suppose that the number of eggs laid by a certain insect has a Poisson distribution with mean λ . The probability that any one egg hatches is p . Assume that the eggs hatch independently of one another. Find the A. Expected value of Y , the total number of eggs that hatch. B. Variance of Y

Let X be the number of eggs laid by an insect and let Y be the number of eggs hatched.

Given that X eggs were laid, Y has a binomial distribution with $p = P[\text{eggs hatched}]$.

That is, $Y|X=x$ has a binomial distribution with parameter x and p .

Therefore,

$$E(Y | X = x) = xp$$

Since X has a Poisson distribution with parameter λ , we have

$$E(Y) = E(Xp) = pE(X) = p\lambda$$

Also,

$$\text{Var}(Y | X = x) = xp(1 - p)$$

We have,

$$\begin{aligned} \text{Var}(Y) &= E[\text{Var}(Y | X)] + \text{Var}[E(Y | X)] \\ &= E[Xp(1 - p)] + \text{Var}(Xp) \\ &= p(1 - p)E[X] + p^2\text{Var}(X) \\ &= p(1 - p)\lambda + p^2\lambda && (\text{Since } E(X) = \lambda \text{ and } \text{Var}(X) = \lambda) \\ &= p\lambda \end{aligned}$$